ELECTRICITY GENERATION ANALYSIS DASHBOARD

Comprehensive Power BI Dashboard for Energy Insights

Comprehensive Project Report | Power BI Edition

106.68K
Total Generation (GWh)
FY2000–FY2025

33.28K
Maximum Generation
Peak Value (GWh)

3.03K
Average Generation
FY2000–FY2025

FY2025
Latest Year
Most Recent Data

DATASET

Global Electricity
Generation Dataset
FY2000 to FY2025 (26 Years)

TOOLS USED

Power BI Desktop
DAX (Data Analysis Expressions)
Power Query (Data Transformation)

DATA SCIENCE MINOR PROJECT REPORT

(Project Semester: January – April 2026)

Electricity Generation Analysis

Using POWER BI

Submitted by

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Registration No.: 12418926

Programme and Section: B.tech CSE, 324MK

Course Code: INT387

Under the Guidance of

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Discipline of CSE/IT

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Lovely Professional University, Phagwara

DECLARATION

I, Jyotirmay Sahoo, student of B.tech CSE under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

This project titled "Electricity Generation Analysis Using Power BI" has been independently carried out and represents original work undertaken during the Project Semester (January–April 2026). The work embodies my own investigation, research, and analysis of publicly available financial data of Tata Motors Limited.

The work embodied in this report has not been submitted to any other University or Institution for the award of any degree or diploma, and all sources used for reference have been duly cited.

I further declare that the Power BI dashboard developed as part of this project is my original creation, and no part of it has been copied from any previously published source without appropriate acknowledgment.

Date: 21-04-2026

Signature of the Student

Place: Phagwara

Registration No.: 12418926

Name: Jyotirmay Sahoo

CERTIFICATE

This is to certify that Jyotirmay Sahoo bearing Registration No.12418926 has successfully completed the INT387 Project titled,

" Electricity Generation Analysis

Using Power BI"

under my guidance and supervision during the Project Semester (January–April 2026) at Lovely Professional University, Phagwara, Punjab. To the best of my knowledge, the present work is the result of her original development, effort, and study.

This project demonstrates the student's ability to apply data science tools and techniques — specifically Microsoft Power BI — to real-world financial data, resulting in an interactive dashboard and a comprehensive analytical report. The student has shown commendable initiative, analytical thinking, and technical proficiency throughout this project.

I recommend this report for evaluation and final submission.

Date: 21-04-2026

_____

Signature and Name of the Supervisor

Designation: Assistant Professor

School of Computer Science and Engineering

Lovely Professional University, Phagwara

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Electricity Generation Analysis.

Lastly, I would like to thank my family and friends for their continuous moral support and encouragement throughout this journey.

[Jyotirmay Sahoo]

ABSTRACT

This project presents a comprehensive analysis of global electricity consumption using Microsoft Power BI as a business intelligence and data visualization tool. The primary objective is to examine electricity consumption patterns across different countries, regions, energy sources, and time periods to identify trends, variations, and key insights.

The dataset used in this project was obtained from publicly available sources and includes information on electricity consumption, generation sources, countries, and yearly data. Data preprocessing was carried out using Power Query, which involved data cleaning, transformation, and structuring to ensure accuracy and consistency.

DAX measures were developed to calculate key performance indicators (KPIs) such as total electricity consumption, average consumption, maximum consumption, and yearly growth trends. An interactive dashboard was designed incorporating KPI cards, bar charts, line charts, pie charts, and map visualizations to effectively represent global electricity consumption patterns.

The analysis highlights significant variations in electricity usage across different regions, identifies high-consumption countries, and reveals trends in energy usage over time. It also provides insights into the contribution of different energy sources such as fossil fuels and renewable energy.

This project demonstrates how Power BI can transform raw energy data into meaningful and actionable insights, enabling better understanding and decision-making in the field of energy management and policy planning.

Keywords: Electricity Consumption, Power BI, Data Visualization, Energy Analysis, Dashboard, Business Intelligence

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1. INTRODUCTION

1.1 Background

Electricity is one of the most essential components of modern economic and social development. It powers industries, transportation systems, households, and digital infrastructure. With the rapid growth of global population and industrialization, electricity consumption has increased significantly over the years. Understanding electricity consumption patterns is crucial for policymakers, energy planners, and stakeholders to ensure sustainable energy management and efficient resource utilization.

The advancement of data analytics and business intelligence tools has made it easier to analyze large-scale energy data. Microsoft Power BI provides powerful capabilities for transforming raw data into meaningful visual insights, enabling users to explore trends, compare regions, and make informed decisions. This project leverages Power BI to analyze global electricity consumption and uncover patterns across countries, regions, and time.

1.2 Overview of Global Electricity Consumption

Global electricity consumption varies widely across countries and regions due to differences in economic development, industrial activity, population size, and energy resources. Developed countries generally exhibit higher electricity consumption due to industrialization and advanced infrastructure, while developing countries show rapid growth due to expanding economies.

Electricity is generated from multiple sources such as fossil fuels (coal, oil, gas), nuclear energy, and renewable sources (solar, wind, hydro). The transition toward renewable energy is a major global trend aimed at reducing environmental impact and ensuring sustainability. Analyzing electricity consumption data helps in understanding these transitions and identifying key contributors to global energy demand

1.3 Objective of the Project

- The main objectives of this project are:

To analyze global electricity consumption across different countries and regions.
To identify trends and patterns in electricity usage over time.
To compare electricity consumption across different energy sources.
To develop interactive dashboards using Power BI for effective data visualization.

- To generate meaningful insights that support decision-making in energy planning.sustainability. Analyzing electricity consumption data helps in understanding these transitions and identifying key contributors to global energy demand.

1.4 Scope of the Project

- The scope of this project includes the analysis of electricity consumption data across multiple countries and years. The project focuses on:
- Country-wise and region-wise electricity consumption
- Year-wise trends in electricity usage
- Comparison of different energy categories
- Visualization of data using Power BI dashboards
- The project is limited to historical data analysis and does not include predictive modeling or real-time data integration.

1.5 Tools and Technologies Used

Tool / Technology	Purpose
Microsoft Power BI	Data visualization and dashboard creation
Power Query	Data cleaning and transformation
DAX	Creating calculated measures and KPIs
CSV Dataset	Source of electricity data

2. SOURCE OF DATASET

2.1 Data Origin

The dataset used in this project was obtained from publicly available sources that provide global electricity consumption data. These sources compile energy-related data from government reports, international organizations, and research institutions. The use of publicly available data ensures transparency and reliability.

2.2 Dataset Description

The dataset consists of multiple attributes related to electricity consumption:

- **Area (Country/Region)**
- **Year**
- **Category (Type of electricity)**
- **Subcategory (Detailed classification)**
- **Continent**
- **Value (Electricity consumption)**
- **Year-over-Year Change**

These attributes allow for multidimensional analysis of electricity consumption patterns.

2.3 Data Size and Format

- Number of records: Multiple rows representing different countries and years
- Number of columns: Approximately 6–7 variables
- File format: CSV
- Data type: Structured dataset

The dataset is suitable for visualization and analysis using Power BI.

3. DATASET PREPROCESSING

3.1 Data Import

The dataset was imported into Power BI using the “Get Data” feature. The CSV file was loaded into the Power Query Editor for further processing.

3.2 Data Cleaning Steps

The following steps were performed:

- Checked for missing values and inconsistencies
- Removed duplicate records
- Standardized column names
- Corrected data types (Year as numeric/date, Value as numeric)

3.3 Data Transformation

Data transformation included:

- Creating calculated columns
- Aggregating data for analysis
- Formatting values for better readability

DAX measures were created for KPIs such as total, average, maximum, and minimum consumption.

3.4 Data Model

The dataset follows a simple structure suitable for analysis. Relationships between fields such as Year and Area allow effective filtering and visualization. The Power BI data model ensures efficient querying and fast performance.

4. ANALYSIS ON DATASET 4. ANALYSIS ON DATASET

4.1 General Description

The analysis focuses on understanding electricity consumption patterns globally. The dashboard provides insights into how electricity usage varies across regions, countries, and time.

4.2 Country-wise Electricity Consumption Analysis

The bar chart visualization shows electricity consumption across different countries. It highlights major contributors to global electricity usage and helps identify high-consumption regions.

4.3 Year-wise Trend Analysis

The line chart displays electricity consumption trends over time. It reveals growth patterns, fluctuations, and changes in electricity demand across years.

4.4 Source-wise Consumption Analysis

The pie chart illustrates the contribution of different energy sources. It helps in understanding the balance between renewable and non-renewable energy usage.

4.5 Dashboard Visualization

The Power BI dashboard includes:

- KPI Cards (Total, Average, Max, Min)
- Bar Charts (Country-wise analysis)
- Line Charts (Trend analysis)
- Pie Charts (Category distribution)
- Maps (Regional analysis)

The dashboard is interactive and allows filtering using slicers.

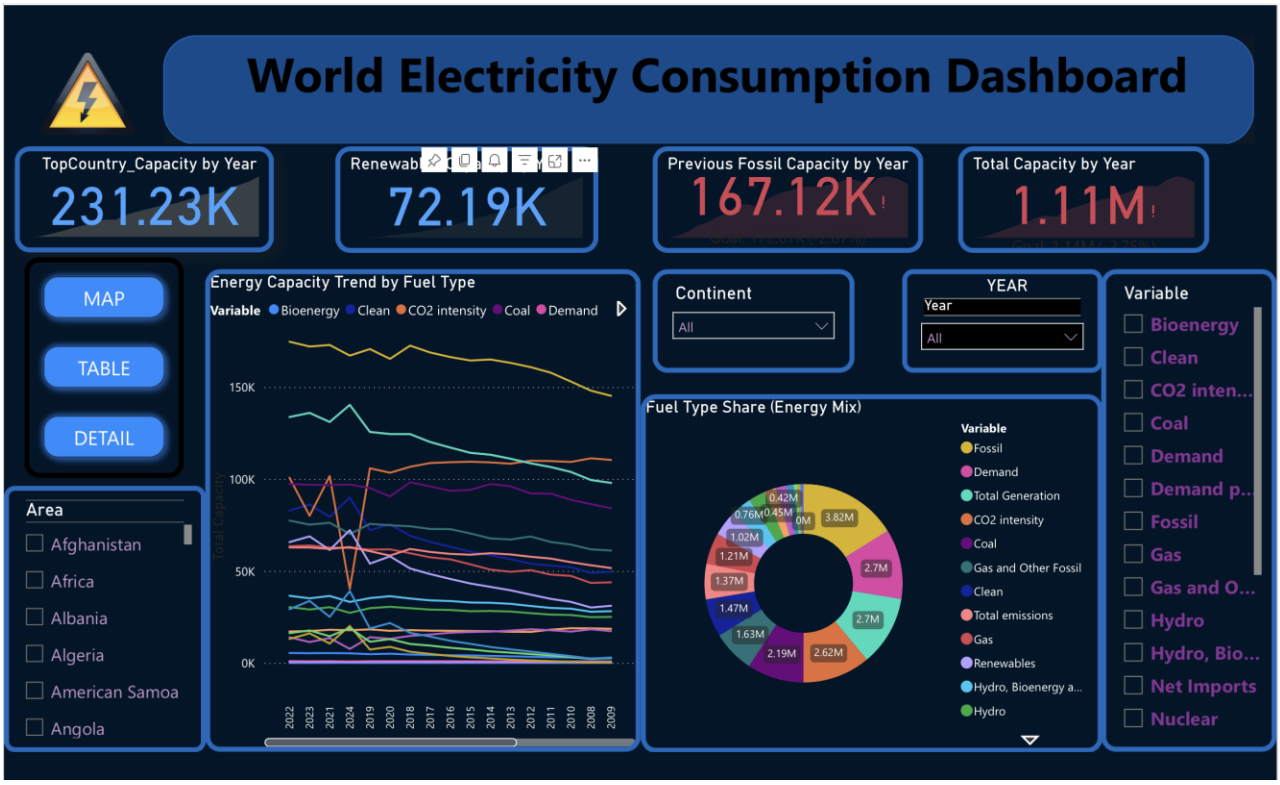

Figure 1: Complete Financial Analysis Dashboard – T

World Electricity Consumption

5. CONCLUSION

This project successfully achieved its objective of analyzing global electricity consumption using Microsoft Power BI as a business intelligence and data visualization tool. Through systematic data collection, preprocessing, transformation, and visualization, raw electricity consumption data was converted into meaningful and actionable insights.

The developed Power BI dashboard provides a comprehensive overview of electricity consumption across different countries, regions, energy categories, and time periods. By integrating multiple visualizations such as KPI cards, bar charts, line charts, pie charts, and maps, the dashboard enables users to explore complex data in an intuitive and interactive manner. The use of slicers and filters further enhances user experience by allowing dynamic exploration of specific regions, years, and energy categories.

One of the key findings of this project is the significant variation in electricity consumption across different countries and regions. Developed countries and industrialized regions tend to have higher electricity consumption due to advanced infrastructure, technological adoption, and industrial activities. In contrast, developing regions show varying patterns of consumption, often influenced by economic growth, population expansion, and access to energy resources.

The year-wise analysis highlights fluctuations and trends in electricity consumption over time. These variations can be attributed to multiple factors such as economic growth, technological advancements, policy changes, and global events. In recent years, there has been a noticeable shift towards sustainable energy usage, reflecting global efforts to reduce dependence on fossil fuels and promote renewable energy sources.

The category-wise analysis provides insights into the distribution of electricity generation sources. It highlights the contribution of different energy types and emphasizes the growing importance of renewable energy such as solar, wind, and hydroelectric power. This shift indicates a global transition towards cleaner and more sustainable energy systems, driven by environmental concerns and international climate goals.

Another important outcome of this project is the demonstration of Power BI as an effective tool for data analysis and visualization. The platform enables users to handle large datasets, perform complex calculations using DAX, and present insights through visually appealing dashboards. The interactive nature of Power BI allows stakeholders to drill down into specific data points and gain deeper insights without requiring advanced technical knowledge.

The project also emphasizes the importance of data-driven decision-making in the energy sector. By analyzing electricity consumption patterns, governments and organizations can make informed decisions related to energy planning, resource allocation, infrastructure development, and policy formulation. The insights derived from this dashboard can support strategic initiatives aimed at improving energy efficiency, reducing environmental impact, and ensuring sustainable growth.

In conclusion, this project demonstrates that business intelligence tools like Power BI can play a crucial role in transforming raw data into valuable insights. The World Electricity Consumption Dashboard provides a clear and structured view of global energy patterns, making it a useful tool for analysts, policymakers, and researchers. The project not only enhances understanding of electricity consumption trends but also highlights the potential of data visualization in solving real-world problems.

6. FUTURE SCOPE

While the current project provides a comprehensive analysis of global electricity consumption using historical data, there are several opportunities to extend and enhance the scope of this work to make it more advanced, dynamic, and impactful.

One of the major improvements can be the integration of **real-time data sources**. Currently, the dashboard is based on static datasets; however, by connecting Power BI to APIs or live data streams from energy agencies and global databases, the dashboard can be transformed into a real-time monitoring system. This would enable users to track electricity consumption trends as they evolve and make timely, data-driven decisions.

Another significant enhancement would be the incorporation of **predictive analytics and forecasting models**. Using time-series analysis techniques or machine learning algorithms such as ARIMA, Prophet, or regression models, future electricity consumption trends can be predicted. Power BI also offers built-in forecasting features that can be leveraged to estimate future demand. This would be highly beneficial for policymakers, energy planners, and utility providers in anticipating future energy requirements.

The project can also be extended by performing a deeper analysis of **renewable versus non-renewable energy sources**. As the global focus shifts towards sustainability, a detailed breakdown of electricity generation by energy type—such as solar, wind, hydro, coal, and natural gas—can provide valuable insights into the transition toward cleaner energy. This would help assess the progress of different countries in adopting renewable energy solutions.

Another important direction for future work is **comparative analysis between countries or regions**. By benchmarking electricity consumption patterns across developed and developing nations, the dashboard can highlight disparities, growth opportunities, and best practices. This comparative study can be further enhanced by incorporating economic indicators such as GDP, population, and industrial output to establish correlations between energy consumption and economic development.

The project can also benefit from integrating **additional datasets** such as carbon emissions, energy efficiency metrics, and climate-related data. This would allow for a more holistic analysis of the environmental impact of electricity consumption and support sustainability-focused decision-making.

Furthermore, the dashboard can be optimized for **mobile devices and responsive layouts**, ensuring accessibility for users on smartphones and tablets. Power BI provides mobile view customization, which can be utilized to create a user-friendly interface for on-the-go analysis.

Another valuable enhancement would be the inclusion of **Natural Language Query (Q&A) features** in Power BI. This would allow users to interact with the dashboard using simple questions like “Which country has the highest electricity consumption?” or “Show trend of renewable energy over time,” making the system more intuitive and accessible for non-technical users.

Automation can also be introduced through **Power BI Service and Power Automate**, enabling scheduled data refresh and automatic report generation. This would eliminate manual updates and ensure that stakeholders always have access to the latest insights.

Lastly, the project can be expanded into a **comprehensive energy intelligence platform** by incorporating multiple layers of analysis, including supply-demand forecasting, energy pricing,

infrastructure planning, and policy evaluation. Such an advanced system would provide a 360-degree view of the global energy ecosystem.

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